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(57) Abstract :

ABSTRACT SMART MILK QUALITY TESTING KIT The present invention relates to a Smart milk quality testing kit (30) that use the biosensor technology to evaluate the quality of the milk in the real time. The present invention provides the he devices is designed to detect the milk with these parameters such as the pH level, temperature, fat content, gas concentration that helps to indicate the milk spoilage or contamination level. The system has the Arduino based controller, a set of sensors including a pH sensor (20), gas sensor (21), FAT sensor module (22) and temperature sensor (23) all enclosed with the portable device. Milk is sample is suck by the milk inlet (16) and guided through a sensor chamber, with the result are displayed on the OLED screen (12). Visual indicators such as the green LED (11) and red LED (18) provide the result of the milk safety. The device is operating on a researchable lithium battery (19) and is equipped with a C-type charging port (17) for portability. A push button (13) allows for sample intake, while power (14) and reset (15) is manage to operate the system. A clock module (25) timestamps test result for traceability. The present invention provides the efficient, affordable, and food safety solution for the user in field of the diary product quality to eliminate the laboratory need to testing milk with multi parameter quality check in a product. Fig 1

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